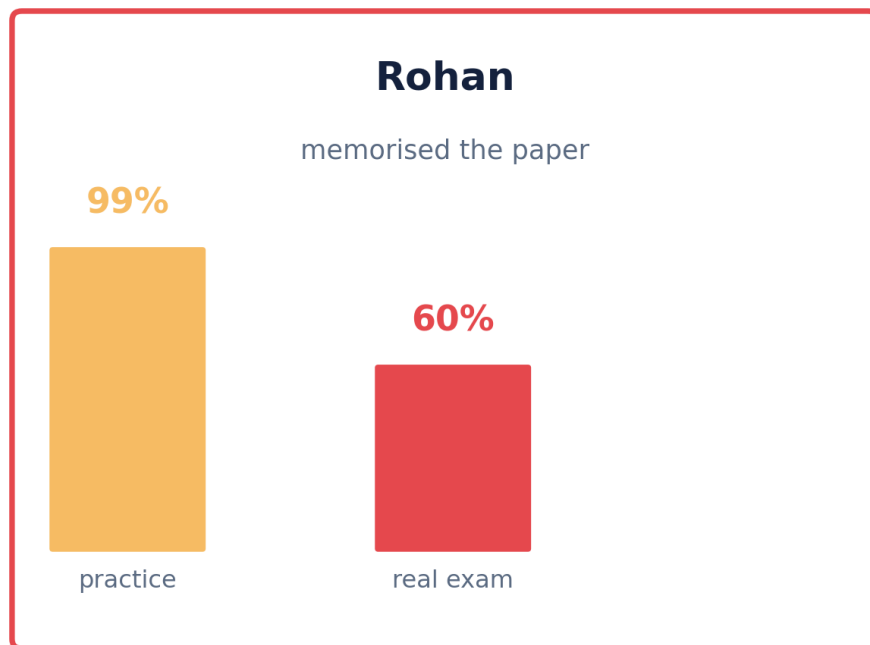
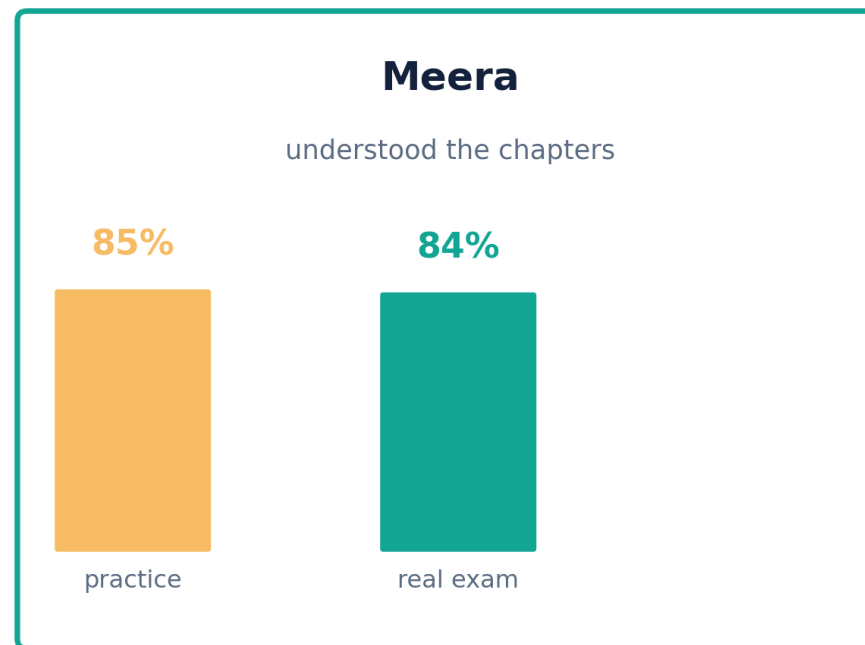


Overfitting, in 90 seconds

SAME PRACTICE TEST · DIFFERENT REAL EXAM



drops 39 points → OVERFIT



holds steady → LEARNED

The GAP between the two bars is overfitting

Rohan memorised the paper. Meera learned the subject. Only one survives new questions.

A model that scores 99% in practice and 60% in the real exam did not learn. It memorised.

- 1 **Rohan** memorised last year's question paper, word for word. **Meera** understood the chapters. Practice test: Rohan 99%, Meera 85%.
- 2 The real exam has *new* questions. Rohan drops to 60%. Meera scores 84%. Rohan learned the noise, not the pattern.
- 3 Spot it with one number: the **gap** between training accuracy and test accuracy. 99% and 60% is not a bad day — it is a memoriser being found out.

REMEMBER

Always hold back test data the model has never seen. It is the only honest exam you can give it.

